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A Naturally Elicited Multimodal Stress Database and Speech Breathing Based Stress Detection

Karumannil Mohamed Ismail Yasar Arafath[1], *Mohammed Abeer K. C.*[2], *Aurobinda Routray*[1]

[1]Department of Electrical Engineering, Indian Institute of Technology, Kharagpur, India
[2]Counselling Centre, Indian Institute of Technology, Kharagpur, India
kmiyasar@gmail.com,mabeer@adm.iitkgp.ac.in, aroutray@ee.iitkgp.ac.in

Abstract

Stress is a natural physiological and psychological response to demanding situations, but accurate detection remains challenging in real-world scenarios. Existing stress databases are limited and often recorded under extreme conditions or acted, which makes real-world applications difficult. This study introduces a multi-modal stress database, naturally elicited through student evaluations, incorporating normal and thermal video alongside speech recordings annotated by psychologists. While physiological signals provide robust stress detection, speech and facial expressions are more accessible. We extract physiological parameters from speech to bridge this gap by automatically detecting breath locations. These breath indices are then used to derive speech breathing parameters. Machine learning techniques classify these parameters for stress detection. Our proposed method marginally outperforms conventional speech-based features, demonstrating a promising approach to stress analysis.

Index Terms: stress, anxiety, stress database, speech breathing, breath detection.

1. Introduction

Anxiety is an unpleasant emotional state characterized by feelings of fear, nervousness, and discomfort, often occurring even in the absence of a real threat [1]. According to the World Health Organization (WHO), in 2015, approximately 3.6% of the global population suffered from anxiety disorders [2]. Despite its prevalence, only 36.9% of individuals with anxiety receive treatment, as reported by the Anxiety and Depression Association of America (ADAA) [3]. A significant barrier to seeking help is the fear and stigma associated with mental illness [4], as individuals worry about its impact on their social and professional lives. This challenge has motivated researchers to explore more accessible and non-invasive methods for detecting stress, particularly in high-risk populations such as students [5].

Stress and anxiety trigger the body's fight-or-flight response, leading to the release of stress hormones like adrenaline and cortisol. These physiological changes manifest as increased heart rate, muscle tension, and altered breathing patterns, including rapid, shallow breathing (hyperventilation) [6]. Hyperventilation disrupts blood carbon dioxide levels, causing symptoms such as dizziness, lightheadedness, and tingling sensations [7]. Research by Borkovec and Costello highlights the significant impact of these respiratory changes on cognitive and emotional processes, ultimately intensifying anxiety [8].

Breathing regulation is crucial in stress management [9]. In a relaxed state, individuals tend to engage in diaphragmatic (deep) breathing, which enhances oxygen exchange and promotes calmness [10]. Under stress, however, breathing shifts to a thoracic (chest) pattern, which is less efficient and can exacerbate anxiety [11]. Studies demonstrate that diaphragmatic breathing exercises effectively reduce anxiety levels [12], while mindfulness-based breathing techniques improve emotional regulation and stress resilience, particularly in students [13].

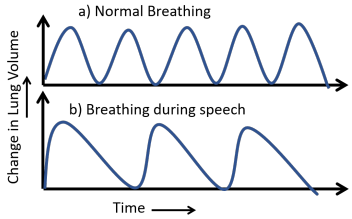

Figure 1: *a) Normal Breathing b) Breathing during speech*

Breath and speech are closely intertwined, as effective speech production relies on controlled breathing. When stress alters breathing patterns, it inevitably affects speech. Speech breathing, the coordination of respiration during speaking, differs from normal breathing by prolonging exhalation. Figure 1 illustrates the contrast between normal and speech breathing.

Stress influences physiological and behavioral responses, including speech, facial expressions, and blood flow. Among these, speech is particularly valuable for stress detection due to its ease of recording. Several speech features have been explored for stress recognition, including fundamental frequency [14], mel-frequency cepstral coefficients (MFCCs) [15, 16], gammatone frequency cepstral coefficients [16], and Teager energy operator (TEO) [17]. The extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) feature set, consisting of 88 parameters, is commonly used for affective computing [18].

Physiological markers such as heart rate, respiration, and skin conductance provide the most reliable means of stress detection. Since speech is inherently linked to breathing, this study aims to extract speech breathing parameters from recorded speech and utilize them for stress detection. The prediction of physiological signals from speech remains a relatively underexplored area [19]. However, existing research has shown that respiratory sinus arrhythmia [19], heart rate [20, 21], and skin conductance [22] can be inferred from speech. Additionally, breath analysis from audio recordings has gained traction, with studies utilizing tracheal sounds [23] and nasal recordings [24] to identify breathing phases. Recent advancements in detecting breath sounds in speech [25] have opened new possibilities despite challenges posed by acoustic similarities to unvoiced phonemes such as fricatives. In this paper, we leverage these techniques to extract speech breathing parame-

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ters for stress detection.

The use of speech breathing for anxiety detection has been explored in early studies. Goldman-Eisler [26] manually evaluated speech breathing metrics such as respiratory rate, speech rate, and ventilation percentage to assess affect during interviews. Kotsopoulos and Clivemellorf [27] employed speech breathing parameters to identify conduct and anxiety disorders in children by manually analyzing breath sounds in speech recordings. Heim et al. [28] integrated chest pneumographs with speech analysis for emotion recognition. Traditionally, speech breathing has been measured using specialized equipment or manual annotation. Still, recent advances in breath detection from speech recordings enable automated speech breathing analysis through speech signal processing and thermal imaging. In the proposed work, we use the breath sound detection in [25] to detect stress.

Despite the importance of stress recognition, it remains underrepresented in existing emotion databases. As of 2003, only two out of 32 databases included stress as an emotion [29]. Some databases, such as [15, 30, 31], contain acted stress responses, while the widely used SUSAS database [32] includes both acted and spontaneous stress. In non-acted databases, stress is either externally induced or naturally occurring. Experimental stress induction methods vary; for instance, SUSAS captures pilot speech under actual and simulated flight conditions, while Luig et al. [33] record speech from pilots in simulated aircraft settings. The utterances are small and hence are not suitable for breath-based study. Other datasets, such as [14], analyze emergency call center interactions for stress assessment. However, as Sigmund [34] pointed out, extreme stress conditions in these databases rarely occur daily. To bridge this gap, our study introduces a naturally elicited stress database, where student evaluations serve as stressors.

The key contributions of this study are: 1) A multi-modal stress database featuring naturally elicited stress during student evaluations, incorporating diaphragmatic breathing for recording a neutral/relaxed baseline. 2) The investigation of speech breathing parameters for stress detection, leveraging recent advancements in speech signal processing.

2. Dataset

The dataset employed in this study is a multimodal, naturally induced stress database focusing on students. The stressor is the evaluation process, as students typically experience stress during assessments. The subjects were internship students at the Indian Institute of Technology (IIT) Kharagpur, India. Data collection took place over two phases, in 2018 and 2019. The dataset includes speech recordings, normal video, and thermal video recordings of students in various scenarios. To preserve the authenticity of the students' emotional responses, chest pneumographs were not used for breath detection, as this method could interfere with natural emotions [26].

2.1. Data collection

The experiment was carried out naturally by embedding all scenarios within the context of an internship or university campus life. The experimental protocol consisted of three distinct phases for data recording: 1) self-introduction, 2) evaluation of the student internship, and 3) delivery of celebration messages.

In the self-introduction phase, subjects are asked to introduce themselves and recite a passage twice. Diaphragmatic breathing is performed to relax the participant before reading the passage for the second time to get a relaxed recording of the subject. Students were expected to be neutral during the self-introduction phase, but the subjects were often stressed.

Table 1: *Summary of dataset statistics*

Description	Count
General Statistics	
Total participants	42
Participants included in database	41
Male participants	26
Female participants	15
Subjects who participated in all phases	26
Subjects with both stressed and neutral recordings	31
Speech recordings extracted from all subjects	176
Emotion Distribution	
Stress recordings	96
Neutral recordings	80
Relaxed recordings	17
Extreme stress cases	6
Participation Across Phases	
Self-introduction participants	37
Stressed during self-introduction	29
Evaluation participants	32
Stressed during evaluation	28
Celebration participants	28
Stressed during celebration	13
Stressed during celebration (Part 2)	10

The evaluation of the work is a critical aspect of the internship, often causing anxiety or stress for the students. During the evaluation, the subject is asked to read the passage when the professor finds the student stressed. To capture the anticipatory anxiety, the subject was asked to read the passage before the evaluation started in the experiment conducted in 2019.

During the conveyance of celebration messages, the participants were asked to read two passages in the context of sending the message to a colleague. The subjects were asked to read their message or a message already given. The subjects were asked to read a passage as in the other two experiment phases.

Speech data were captured with an Ahuja headband microphone (HBM-50) and Audacity, sampled at 44.1 kHz. Thermal footage was shot on a Fluke Ti400 infrared camera at 9 fps, and video was recorded on a Nikon DSLR at 59.94 fps. All sessions took place in an air-conditioned, acoustically isolated room to control temperature and suppress ambient noise. Every participant provided written informed consent for recording and research use, and the protocol was approved by the Institutional Ethics Committee of IIT Kharagpur.

This study utilizes recordings exclusively from the passages read during the three phases. Specifically, two recordings come from the self-introduction and celebration phases each, while the evaluation phase includes one or two recordings per participant. The dataset comprises 41 participants, 26 of whom are male, totaling 176 audio recordings of spoken passages with an average length of 20.25 seconds.

2.2. Data annotation

The students have been asked to fill out the short-STAI [35] to know the state anxiety of the students. However, it is challenging to accurately assess the anxiety/stress levels of subjects during different stages, such as self-introduction and passage reading. To address this, two professional psychologists

Figure 2: *Proposed block diagram for the anxiety/stress detection*

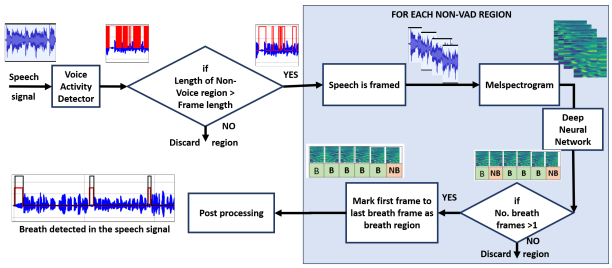

Figure 3: *Block diagram for breath sound detection*

with 15 years of clinical and counseling experience with college students were asked to perform the annotations. They are employed as mental health counselors at IIT Kharagpur. They independently assessed the students' stress levels using normal video recordings with speech, rating them from relaxed, neutral, mildly anxious, anxious, to very anxious, mapped to a numerical scale of 0 to 4. The average of their ratings was calculated, with values less than two categorized as neutral.

The psychologists utilized various biomarkers to assess stress levels in the students, including lip bites, lip wipe rate, neck tilt, saccadic eye movement, eyebrow movements, and non-enjoyment smiles. The inter-rater agreement was evaluated using Cohen's Kappa coefficient, which was found to be 0.812, indicating good agreement between the two raters. The dataset consists of 96 stressed and 80 neutral speech recordings.

3. Methodology

The proposed block diagram of the stress detection from speech using speech-breathing is shown in Figure 2. This work involves two main tasks: 1) the detection of breath sounds in speech, 2) extraction of the speech breathing features, and 3) stress detection from the speech breathing features derived from the breath sound locations.

3.1. Breath sound detection

The breath sound detection has been performed in our earlier work [25], where we propose a deep-learning pipeline for isolating breath sounds within continuous speech. The system first applies a voice-activity detector (VAD) to mark unvoiced intervals, which are then split into short frames. Each frame is converted to a mel-spectrogram and passed through a bidirectional LSTM that labels it as either "breath" or "non-breath." Within every VAD-defined unvoiced segment, frames tagged as breaths are grouped: if at least two such frames are present, the stretch from the first to the last is taken as a single breath event. A post-processing stage discards detections shorter than roughly 100 ms. An overview of the workflow appears in Figure 3.

3.2. Speech Breathing features

The output of the breath sound detection is the start and end time for each breath sound detected. The speech breathing parameters are calculated based on these parameters. Figure 4

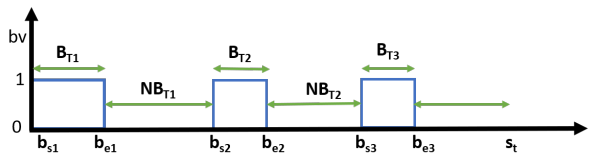

Figure 4: *Breath parameters in speech*

shows a sample breath sound detection in the speech. In the figure, b_s and b_e denote the time indices of the breath. The breath vector value 1 on y axis indicates breath detection. The total speech duration is indicated by s_t. The breath and inter-breath intervals are indicated B_T and NB_T. The following speech breathing parameters are used in this work.

3.2.1. *Temporal features*

The temporal features are the breath duration (B_T) and inter-breath interval (NB_T).

3.2.2. *Energy features*

The energy of the breath (B_T) and non-breath (NB_T) region is found by the root mean square of the speech and breath waveform.

3.2.3. *Frequency features*

Respiratory rate (RR) and energy per breath (EPB) are frequency features and are calculated using equations 1 and 2.

$$RR = \frac{\text{No. of breaths}}{\text{Total duration}} \quad (1)$$

$$EPB = \frac{\text{Total energy}}{\text{No. of breaths}} \quad (2)$$

3.2.4. *Positional Features*

The relative positions of breaths within the speech recordings have been used as the positional features.i.e., b_{s1}, b_{s2},etc.

3.2.5. *Derived features*

Speech is produced using the inhaled air in the lungs. So, the ratio of breath duration to the non-breath interval after the breath and the ratio of the non-breath interval to the breath interval following the non-breath interval have been derived. The energy feature is another feature when considering the amount of air inhaled and used for speech. The consistency of the breath duration and energy is calculated as the ratio of standard deviation to the mean.

The normalized breath duration,inter-breath interval, and their statistical features are included. The percentage of breath time is calculated as total breath duration as a percentage of total speech duration.

3.2.6. *Breath regularity indices*

The rhythmic patterns in the breath are found by taking the autocorrelation of the breaths. It helps to find whether there are any changes in the pattern of breaths during the stress.

3.2.7. *Advanced statistical features*

The skewness and kurtosis calculate the asymmetry and peakedness of the breath and inter-breath duration.

The mean, standard deviation, minimum, and maximum values of temporal, energy, positional, and breath regularity indices are used for the classification.

3.3. Stress detection

There are 61 speech breathing parameters extracted from each speech signal. The extracted parameters were normalized, and Principal Component Analysis (PCA) was applied for dimensionality reduction. Since not all subjects had both stressed and neutral recordings, the subjects were divided into 10 groups, each containing subjects who had both types of recordings. A Leave-One-Group-Out (LOGO) validation strategy was employed to ensure speaker-independent evaluation. Various machine learning classifiers, including Decision Tree, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Random Forest (RF), and XGBoost, were used for classification. The models were implemented in Python using the Scikit-Learn library, with hyperparameter tuning performed via Grid Search. Additionally, a 10-fold cross-validation was conducted to evaluate the model's performance.

4. Results

The dataset analysis revealed significant patterns in stress manifestation across different contexts. Stress was the predominant emotional state, with 96 stress-labeled instances, compared to 80 neutral recordings, including 17 relaxed cases. Contextual influences on anxiety were evident throughout different study phases. During the self-introduction phase, 29 out of 37 participants exhibited stress, reinforcing the well-documented stress response to public speaking. The diaphragmatic exercise reduced the stress in the subjects, and 17 recordings in the relaxed state were obtained. The evaluation phase elicited even stronger stress reactions, with 28 out of 32 participants showing heightened stress levels, underscoring the psychological impact of performance assessments. Interestingly, even in the celebration phase, which is typically associated with positive emotions, 13 participants initially displayed anxiety during the first passage reading. However, this number decreased to 10 in the second reading, suggesting a potential adaptive response over time. The recording environment or public speaking might be the reason for this stress. The recordings were around 20 seconds, which is neither lengthy nor too small. This provides a crucial foundation for investigating stress detection through speech breathing parameters, offering insights into the physiological correlates of stress across various situational contexts.

The breath sound detection algorithm achieved an F1-score of 96% [25]. Speech breathing features were extracted to classify speech signals as stressed or neutral. To ensure subject independence, 10-fold cross-validation was performed, ensuring no subject appeared in training and testing sets. The process was repeated with different data splits, and the average performance metrics were reported. Table 2 summarizes the classification results, with XGBoost achieving the highest performance, an F1-score of 59%, and an accuracy of 64%.

To further analyze the effectiveness of speech breathing features, we compared them with two other commonly used feature sets: eGeMAPS and TEO. The classification performance of these feature sets, evaluated using different machine learning algorithms, is presented in supplementary sheet *results.pdf*. The results indicate that eGeMAPS performed best with the Random Forest classifier, achieving an F1-score of 0.59 and accuracy of 62%, whereas TEO yielded the best results with KNN, with an F1-score of 0.54 and accuracy of 56%.

Table 2: *Performance of speech breathing features on various classifiers.*

Classifier	Precision	Recall	F1	Accuracy
Decision Tree	0.53	0.53	0.50	0.53
KNN	0.56	0.58	0.51	0.56
SVM	0.53	0.55	0.47	0.57
RF	0.57	0.56	0.51	0.59
XGBoost	**0.64**	**0.62**	**0.59**	**0.64**

Table 3: *Comparison of various features for classification.*

Feature	Classifier	Prec.	Recall	F1	Accu.
TEO	KNN	0.56	0.56	0.54	0.56
eGeMAPS	RF	0.62	0.61	0.59	0.62
Speech breathing	**XGBoost**	**0.64**	**0.62**	**0.59**	**0.64**

The best-performing models for each feature set were further compared in Table 3. Speech breathing features with XGBoost consistently outperformed the other feature sets, achieving the highest classification performance with an F1-score of 0.59 and an accuracy of 64%. In contrast, eGeMAPS features with Random Forest achieved an F1-score of 0.59 and an accuracy of 62%, while TEO features using KNN achieved an F1-score of 0.54 and accuracy of 56%. The parameters of the best model is provided in the supplementary sheet.

Overall, these findings demonstrate the effectiveness of speech breathing features for stress classification, achieving marginally better results over traditional feature sets. However, it is important to note that speech breathing-based classification requires detectable breath sounds in speech recordings for accurate stress detection.

Limitations: Despite the promising results, this work has certain limitations. While expert annotation provides valuable insights into stress detection, the latest sensors provide non-contact measurements of physiological parameters for stress detection. Additionally, the effectiveness of speech breathing features relies on the accurate detection of breath sounds. Any errors in breath segmentation may propagate to the extracted features, potentially affecting the overall performance of stress detection.

Future research should explore a broader range of physiological and breathing parameters, such as lung volume changes, to improve detection accuracy.

5. Conclusion

This work proposed a naturally elicited multi-modal stress database. It automates the calculation of speech breathing parameters by detecting the breath sounds in the speech and uses the parameters for stress detection. The speech breathing parameters performed marginally better than TEO and eGeMAPS features. However, a limitation is that the speech recordings must be of sufficient length to capture both breath and speech for the method to be effective.

6. References

[1] L. M. Shin and I. Liberzon, "The neurocircuitry of fear, stress, and anxiety disorders," *Neuropsychopharmacology*, vol. 35, no. 1, pp. 169–191, 2010.

[2] W. H. Organization *et al.*, "Depression and other common mental disorders: global health estimates," *(WHO reference number: WHO/MSD/MER/2017.2)*, 2017.

[3] Anxiety and D. A. of America. Facts and statistics. [Online]. Available: https://adaa.org/about-adaa/press-room/facts-statistics

[4] H. Haritha, S. Negi, R. S. Menon, A. A. Kumar, and C. S. Kumar, "Automating anxiety detection using respiratory signal analysis," in *IEEE Region 10 Symposium (TENSYMP), 2017*. IEEE, 2017, pp. 1–5.

[5] N. Bayram and N. Bilgel, "The prevalence and socio-demographic correlations of depression, anxiety and stress among a group of university students," *Social psychiatry and psychiatric epidemiology*, vol. 43, no. 8, pp. 667–672, 2008.

[6] B. S. McEwen, "Physiology and neurobiology of stress and adaptation: central role of the brain," *Physiological reviews*, vol. 87, no. 3, pp. 873–904, 2007.

[7] G. P. Chrousos and P. W. Gold, "The concepts of stress and stress system disorders: overview of physical and behavioral homeostasis," *Jama*, vol. 267, no. 9, pp. 1244–1252, 1992.

[8] T. Borkovec and E. Costello, "Efficacy of applied relaxation and cognitive-behavioral therapy in the treatment of generalized anxiety disorder." *Journal of consulting and clinical psychology*, vol. 61, no. 4, p. 611, 1993.

[9] R. Jerath, J. W. Edry, V. A. Barnes, and V. Jerath, "Physiology of long pranayamic breathing: neural respiratory elements may provide a mechanism that explains how slow deep breathing shifts the autonomic nervous system," *Medical hypotheses*, vol. 67, no. 3, pp. 566–571, 2006.

[10] R. P. Brown and P. L. Gerbarg, "Sudarshan kriya yogic breathing in the treatment of stress, anxiety, and depression: part i—neurophysiologic model," *Journal of Alternative & Complementary Medicine*, vol. 11, no. 1, pp. 189–201, 2005.

[11] R. Ley, "The modification of breathing behavior: Pavlovian and operant control in emotion and cognition," *Behavior Modification*, vol. 23, no. 3, pp. 441–479, 1999.

[12] Y.-F. Chen, X.-Y. Huang, C.-h. Chien, and J.-F. Cheng, "The effectiveness of diaphragmatic breathing relaxation training for reducing anxiety," *Perspectives in psychiatric care*, vol. 53, no. 4, pp. 329–336, 2017.

[13] Z. Jia-Yuan, J. Xiang-Zi, F. Yi-Nan, and C. Yu-Xia, "Emotion management for college students: effectiveness of a mindfulness-based emotion management intervention on emotional regulation and resilience of college students," *The Journal of Nervous and Mental Disease*, vol. 210, no. 9, pp. 716–722, 2022.

[14] G. Demenko and M. Jastrzebska, "Analysis of voice stress in call centers conversations," in *Speech Prosody 2012*, 2012.

[15] R. Sabo, M. Rusko, A. Ridzik, and J. Rajčáni, "Stress, arousal, and stress detector trained on acted speech database," in *International Conference on Speech and Computer*. Springer, 2016, pp. 675–682.

[16] S. Besbes and Z. Lachiri, "Multi-class svm for stressed speech recognition," in *2016 2nd International Conference on Advanced Technologies for Signal and Image Processing (ATSIP),*. IEEE, 2016, pp. 782–787.

[17] G. Zhou, J. H. Hansen, and J. F. Kaiser, "Nonlinear feature based classification of speech under stress," *IEEE Transactions on speech and audio processing*, vol. 9, no. 3, pp. 201–216, 2001.

[18] F. Eyben, K. R. Scherer, B. W. Schuller, J. Sundberg, E. André, C. Busso, L. Y. Devillers, J. Epps, P. Laukka, S. S. Narayanan *et al.*, "The geneva minimalistic acoustic parameter set (gemaps) for voice research and affective computing," *IEEE Transactions on Affective Computing*, vol. 7, no. 2, pp. 190–202, 2016.

[19] A. Jati, P. G. Williams, B. Baucom, and P. Georgiou, "Towards predicting physiology from speech during stressful conversations: Heart rate and respiratory sinus arrhythmia," in *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, April 2018, pp. 4944–4948.

[20] D. Skopin and S. Baglikov, "Heartbeat feature extraction from vowel speech signal using 2d spectrum representation," in *Proc. the 4th Int. Conf. Information Technology*, 2009.

[21] J. Smith, A. Tsiartas, E. Shriberg, A. Kathol, A. Willoughby, and M. de Zambotti, "Analysis and prediction of heart rate using speech features from natural speech," in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 989–993.

[22] B. Schuller, F. Friedmann, and F. Eyben, "Automatic recognition of physiological parameters in the human voice: Heart rate and skin conductance," in *2013 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2013, pp. 7219–7223.

[23] B. A. Reyes, N. Reljin, Y. Kong, Y. Nam, S. Ha, and K. H. Chon, "Towards the development of a mobile phonopneumogram: automatic breath-phase classification using smartphones," *Annals of biomedical engineering*, vol. 44, no. 9, pp. 2746–2759, 2016.

[24] A. Abushakra and M. Faezipour, "Acoustic signal classification of breathing movements to virtually aid breath regulation," *IEEE Journal of Biomedical and Health Informatics*, vol. 17, no. 2, pp. 493–500, 2013.

[25] K. M. I. Y. Arafath and A. Routray, "Detection of breath sounds in speech: a deep learning approach," *Engineering Applications of Artificial Intelligence(under review)*, http://dx.doi.org/10.2139/ssrn.4921332.

[26] F. Goldman-Eisler, "Speech-breathing activity—a measure of tension and affect during interviews," *British Journal of Psychology*, vol. 46, no. 1, pp. 53–63, 1955.

[27] S. Kotsopoulos and C. Mellor, "Extralinguistic speech characteristics of children with conduct and anxiety disorders," *Journal of Child Psychology and Psychiatry*, vol. 27, no. 1, pp. 99–108, 1986.

[28] E. Heim, P. H. Knapp, L. Vachon, G. G. Globus, and S. J. Nemetz, "Emotion, breathing and speech," *Journal of Psychosomatic Research*, vol. 12, no. 4, pp. 261–274, 1968.

[29] D. Ververidis and C. Kotropoulos, "A review of emotional speech databases," in *Proc. Panhellenic Conference on Informatics (PCI)*, 2003, pp. 560–574.

[30] I. Lefter, G. J. Burghouts, and L. J. Rothkrantz, "An audio-visual dataset of human–human interactions in stressful situations," *Journal on Multimodal User Interfaces*, vol. 8, no. 1, pp. 29–41, 2014.

[31] F. Burkhardt, A. Paeschke, M. Rolfes, W. F. Sendlmeier, and B. Weiss, "A database of german emotional speech," in *Ninth European Conference on Speech Communication and Technology*, 2005.

[32] J. H. Hansen and S. E. Bou-Ghazale, "Getting started with susas: A speech under simulated and actual stress database," in *Fifth European Conference on Speech Communication and Technology*, 1997.

[33] J. Luig and A. Sontacchi, "A speech database for stress monitoring in the cockpit," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, vol. 228, no. 2, pp. 284–296, 2014.

[34] M. Sigmund, "Introducing the database examstress for speech under stress," in *Signal Processing Symposium, 2006. NORSIG 2006. Proceedings of the 7th Nordic*. IEEE, 2006, pp. 290–293.

[35] T. M. Marteau and H. Bekker, "The development of a six-item short-form of the state scale of the spielberger state—trait anxiety inventory (stai)," *British Journal of Clinical Psychology*, vol. 31, no. 3, pp. 301–306, 1992.